

Imperial College London
Department of Earth Science and Engineering
MSc in Applied Computational Science and Engineering

Independent Research Project
Project Plan

Solving the Wave Equation Numerically by Optimising a Discrete Loss

by

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repository: <https://github.com/ease-ada-lovelace-2025/irp-ld2022>

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June 2026

Abstract

The numerical modelling of waves has broad applications across science and engineering. There are many ways to represent waves computationally, but each requires discretisation, relying on discretisation schemes like the finite difference. Discretisation introduces errors that can compound over time, particularly when applied to the wave equation. Recently, the Optimising a Discrete Loss (ODIL) framework has been proposed as an efficient method for producing accurate approximations of the solution of Partial Differential Equations (PDE) by means of minimising a discrete loss function. ODIL does not require a time stepping approach, unlike classical numerical methods, which may offer accuracy and efficiency benefits when solving the wave equation. This work aims to explore the accuracy and performance characteristics of the ODIL framework when applied to the wave equation.

Problem Description

Introduction

The modern world depends strongly on our understanding of wave propagation. With applications spanning medical imaging, resource prospecting, and telecommunications, the implications of wave propagation are countless and have been central to the development of civilisation. Supporting this development is our ability to solve the wave equation numerically, that is, to approximate it by a discrete form so that it can be represented on a computer.

The propagation of a wave field u through time t and space can be described by the wave equation,

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u, \quad (1)$$

where c is the wave speed and ∇^2 is the Laplacian operator in one, two, or three spatial dimensions.

Under certain assumptions, an analytical solution to the wave equation can be derived, but in many practical contexts we require the flexibility to represent complex geometries, boundary conditions, and forcing terms at will. To do so, we should look to solve the continuous wave equation numerically by approximating it with a discrete form. Classical discretisation approaches include the finite difference [19, 7] and finite element [20] methods. These result in a system of equations which can be solved iteratively [13], or directly [12], but either approach incurs a significant computational cost due to the often ill-conditioned and sparse nature of the discretised systems (see [18, 15, 11]). Regardless of the solution method, discretisation introduces an error that must be balanced with computational expense.

Other errors arise when solving the wave equation in particular. Numerical dispersion and dissipation errors occur where a finite difference discretisation introduces phase and amplitude inconsistencies [2]. Resultantly, waves can propagate with an incorrect speed, meaning errors accumulate with propagation distance. Additionally, errors tend to compound over long time integrations [1]. Often, a large number of discretisation nodes are required to achieve an acceptable accuracy [12], which contributes to the computational workload. Many advancements have been made to combat these sorts of errors, such as high order [6] and dispersion reduction [10, 3] schemes, and spectral methods [4], but the necessity for robust wave modelling tools motivates a continued search for numerical methods that are both accurate and computationally efficient.

Method overview

Optimising a Discrete Loss (ODIL) [14] has emerged as a novel technique for the numerical solution of PDEs and their associated inverse problems. ODIL approximates the solution to a PDE by means of minimising a functional $\mathcal{L}$ that encapsulates the discretised PDE, boundary, and initial conditions g_i^n . For a 1D wave equation discretised using second order central differences, the ODIL functional

$$\mathcal{L}(u) = \underbrace{\sum_{(i,n) \in \Omega} \left(\frac{u_i^{n+1} - 2u_i^n + u_i^{n-1}}{\Delta t^2} - \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2} \right)^2}_{\text{PDE residual}} + \underbrace{\sum_{(i,n) \in \Gamma} (u_i^n - g_i^n)^2}_{\text{conditions residual}} \quad (2)$$

represents how strongly both the PDE and boundary and initial conditions are satisfied by the current discrete solution field u_i^n (at the n -th temporal node and i -th spatial node) on the interior and boundary domains Ω and Γ . As such, when the loss is minimised, the PDE and associated conditions are satisfied (within some tolerance). Figure 1 presents the result of minimising this loss function.

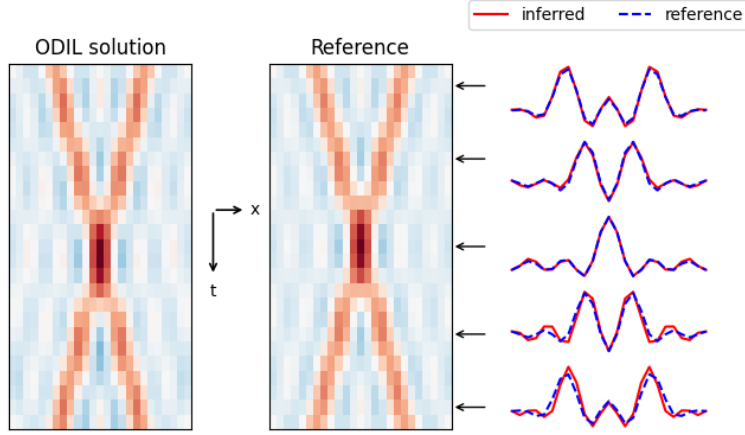


Figure 1: Discrete wave field u_i^n (left) versus a reference solution (middle) obtained by minimising the functional in Equation 2 by the L-BFGS-B method. The wave field is discretised into an $N \times I$ uniform grid, where $N = I = 25$ and $t \in [0, 1]$, $x \in [-1, 1]$. The result and reference employ an analytical solution described in full in [14].

The benefit of formulating the problem in this way is threefold. Firstly, ODIL inherits the accuracy, convergence, and conservation properties of the numerical methods that employ the same grid-based discretisations. Secondly, the optimisation can be orchestrated by the highly optimised automatic differentiation tools of modern machine learning libraries. Thirdly, direct optimisation removes the need for a Marching On in Time (MOT) approach by construction, so the issue of time-compounding errors is avoided. This could be particularly impactful for solving the wave equation, considering it is subject to errors that can accumulate over time [16].

Objectives

The work of [14] demonstrates that ODIL can approximate the solution to PDEs accurately. However, the work does not establish a comparison between the framework and classical numerical methods, nor does it explore the wave equation beyond the first spatial dimension. This work intends to build on the foundations of ODIL by exploring its application for solving the wave equation in two spatial dimensions. In particular, our objectives are

- i To explore how the wave equation can be solved using the ODIL framework in two or more dimensions.
- ii To quantify and analyse the accuracy and runtime performance of ODIL for the solution of the wave equation in comparison to established implementations of finite difference methods.
- iii To develop a robust, benchmarked computational tool that is capable of applying ODIL to problems involving the solution of the wave equation.

Methodology

A timeline for the below methodology is presented in Figure 2.

Solving the 2D Wave Equation with ODIL

The Optimising a Discrete Loss framework casts a numerical PDE solve as an optimisation problem. The discrete PDE and associated conditions are gathered into a functional $\mathcal{L}$ such that when it is minimised, the PDE and conditions are satisfied. The discrete field $\hat{u}$ itself is the target variable, and it can be optimised via a gradient-based method, for example. For a continuous 2D wave field $u := u(t, x, y)$, the wave equation with a forcing term f is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + f. \quad (3)$$

Under ODIL, the discrete functional for this equation takes the form

$$\mathcal{L}(\hat{u}) = \sum_{(i,n) \in \Omega} \left(\frac{\partial^2 \hat{u}}{\partial t^2} - c^2 \left(\frac{\partial^2 \hat{u}}{\partial x^2} + \frac{\partial^2 \hat{u}}{\partial y^2} \right) - f \right)^2 + \sum_{(i,n) \in \Gamma} (\hat{u} - \hat{g})^2, \quad (4)$$

where $\hat{g}$ encodes the associated boundary and initial conditions. Equation 3 and its conditions will be discretised using high order finite difference stencils (represented by sparse matrix operators) and solved by optimising the functional in Equation 4. At every stage, we will verify the correctness of our methods through error and convergence analysis, and the method of manufactured solutions.

Experimentation and Implementation

We will quantify and analyse the performance of the methods we implement. For example, we will assess accuracy using metrics such as pointwise error and global norms. Specifically, we will characterise the dispersion and dissipation error. We will evaluate stability through von Neumann stability analysis [17] and phase analysis [5]. We will benchmark the software against an established finite difference-based solver like `stride` [8]. In scripts and notebooks, we will demonstrate intended usage and how to reproduce figures in the written report.

Some of the key features we intend to implement are summarised in Table 1.

Feature	Purpose
High order finite difference schemes	To enable the user to solve the wave equation accurately.
Sparse matrix operators	To leverage the efficient sparse matrix operations in the <code>scipy.sparse</code> module.
Pytorch wrappers	To leverage existing automatic differentiation and optimisation tools and provide a fully differentiable framework.
Various boundary and initial conditions	To allow the software to be used for modelling tasks across many physical settings.
Optional source term	For applications such as medical and subsurface imaging, the wave equation is typically solved with a known wavelet.
Off grid sources and receivers	To allow for wave field measurements to be taken at an arbitrary spatial position, enabled by sink interpolation between grid nodes.

Table 1: Summary of a selection of the intended features of our software and their purpose.

Extended Study

Conclusions based on the experimentation phase will drive the extended study. It is difficult to say at this point what this will involve, but candidates include

- Expansion to the 3D wave equation.
- Implementation of other wave equations, such as the elastic wave equation.
- Code optimisations.

We might also choose to explore other problems using the framework, such as quantifying uncertainty, and understanding limitations of the algorithm for domain-specific problems such as medical imaging.

AI Acknowledgement Statement

I used Perplexity AI’s Sonar 2 (<https://www.perplexity.ai/>) for exploring the literature on wave modelling, discretisation schemes, and numerical modelling in general. Specifically, I asked for reading recommendations to gain an understanding of the aforementioned fields. I verified the existence, correctness, and relevance of all sources myself. The submitted work reflects my own understanding, design decisions, and technical contributions as declared in the Academic Integrity Declaration.

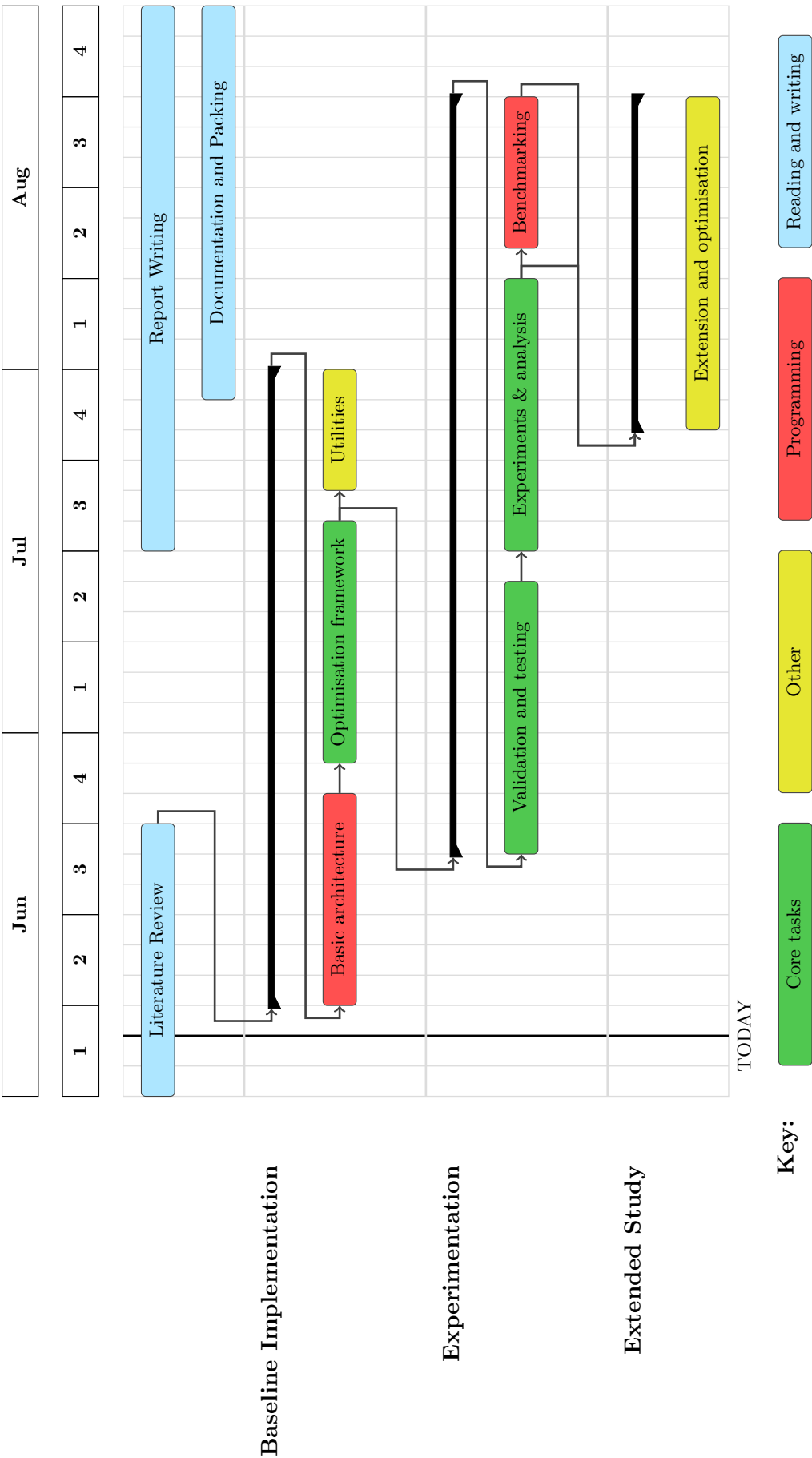


Figure 2: Gantt chart detailing the proposed work spanning the IRP period. Report and documentation writing is expected to be conducted in parallel with other tasks; this figure suggests when the bulk of the writing will take place. Elements of the Gantt chart borrow from the templates by [9].

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